Training runs: ANP with BGE-M3 item embeddings (cross-attention, 1 self-at

Same architecture throughout; the split is the variable. Replay: random acquisition, macro-task Brier ALC.

run	split	width / latent	context sizes, method	steps	best step	bank Brier	replay val	replay test
ANP, random-subject split	random subject	64 / 32	uniform, 0-31 labels	100,000	79,700	0.1248	0.1519	0.2465
ANP, random-subject split (width 128)	random subject	128 / 128	uniform, 0-31 labels	100,000	97,800	0.1193	0.1485	0.2515
ANP	benchmark holdout	64 / 32	uniform, 0-31 labels	100,000	1,200	0.1675	0.1735	0.2725
ANP, budget-ladder training	benchmark holdout	64 / 32	budget ladder, ALC weights	15,000	50	0.2105	not run	not run
ANP + SAM	benchmark holdout	64 / 32	budget ladder, equal weights, SAM 0.05	30,000	5,450	0.2058	0.1667	0.2663
ANP + prior anchor	benchmark holdout	64 / 32	budget ladder, equal weights, anchor	10,000	100	0.2066	not run	not run
ANP + SAM, budget-ladder training	benchmark holdout	64 / 32	budget ladder, ALC weights, SAM 0.05	30,000	4,350	0.2063	not run	not run
ANP, dense-prefix training	benchmark holdout	64 / 32	every prefix, 0-31	10,000	7,800	0.1694	0.1735	0.2762

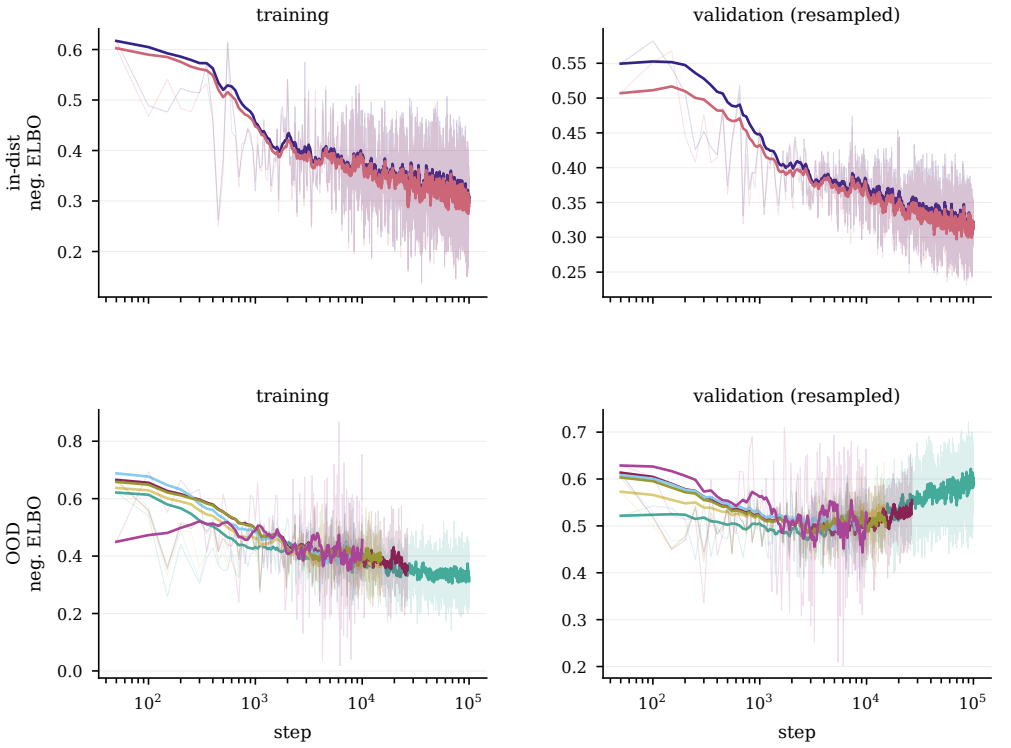
bank Brier: best value on the run's own fixed validation bank, the metric that picked its checkpoint. That bank's context sizes follow the run's sampling scheme, and in-distribution banks hold benchmarks the run trained on, so bank Brier compares only runs with the same bank.

replay val / test: random acquisition on the 82-task validation and 45-task test panels, the same protocol for every row. The random-split runs trained on ai2d_test and bird_sql (validation) and editbench (test).

The dense-prefix run kept training after its replay and CodaBench checkpoint was frozen at step 4,550 (bank Brier 0.1700); its final best is the step shown. SAM ran to 30,000 steps but its best checkpoint, step 5,450, did not change.

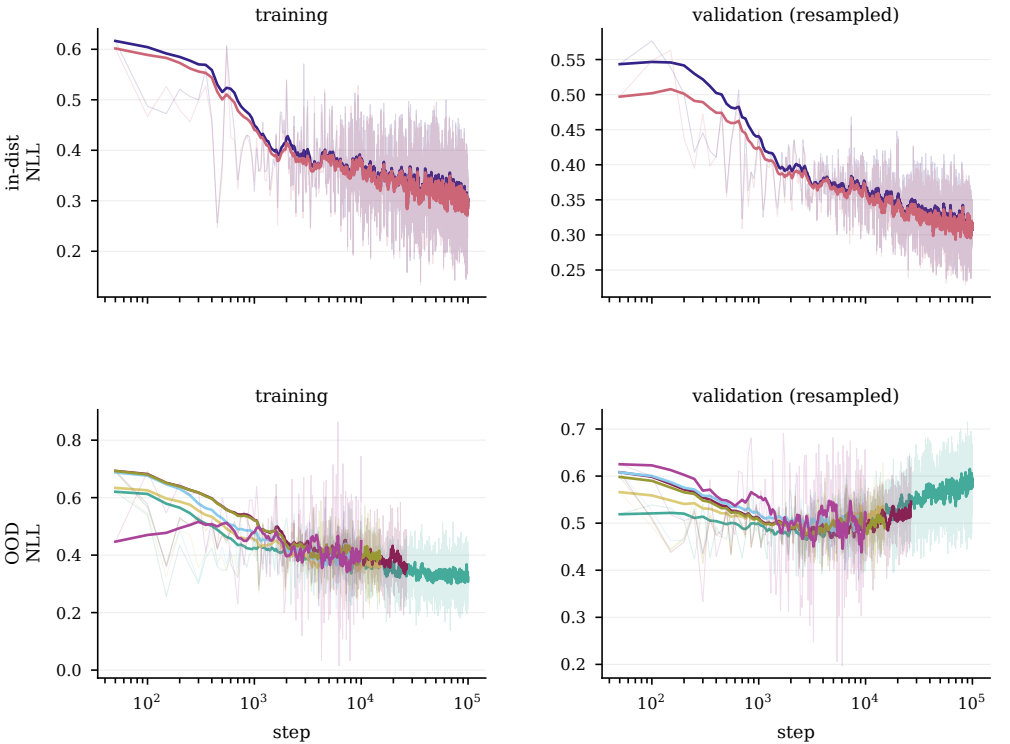
Objective: negative ELBO per target

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



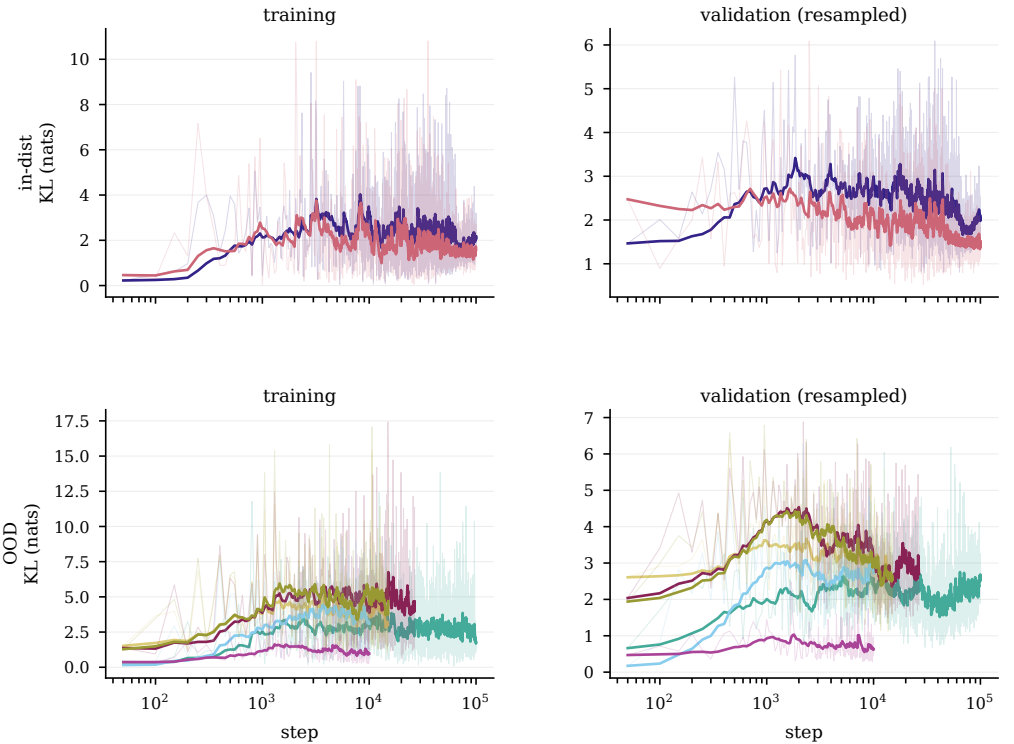
Posterior target NLL

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



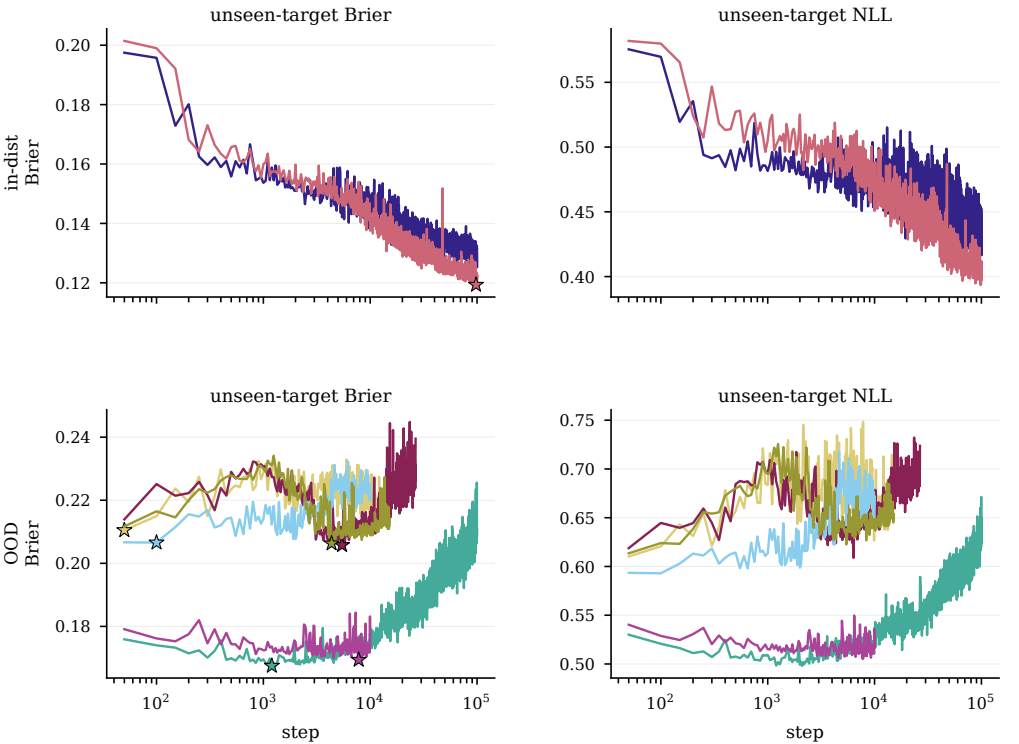
Latent KL (total)

x: optimiser step (log scale). Faint: logged values; solid: exponential moving average (span 20 points).



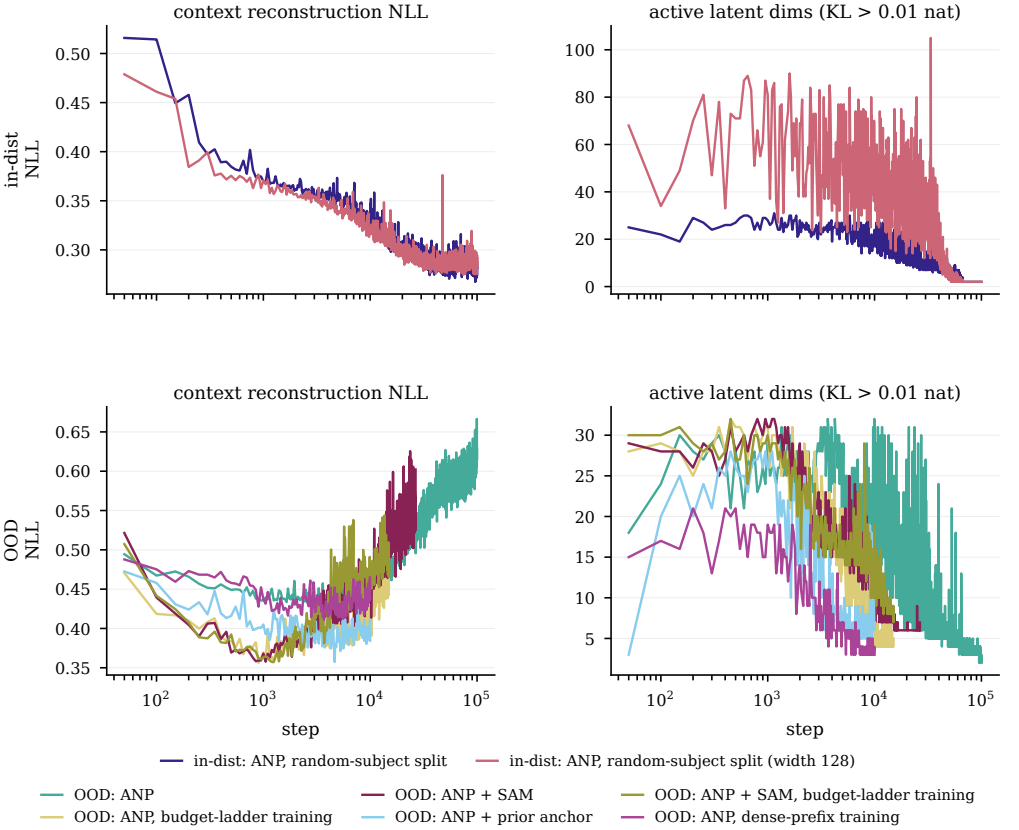
Fixed validation bank: checkpoint-selection metric

x: optimiser step (log scale). Star: the step each run's checkpoint was selected at.



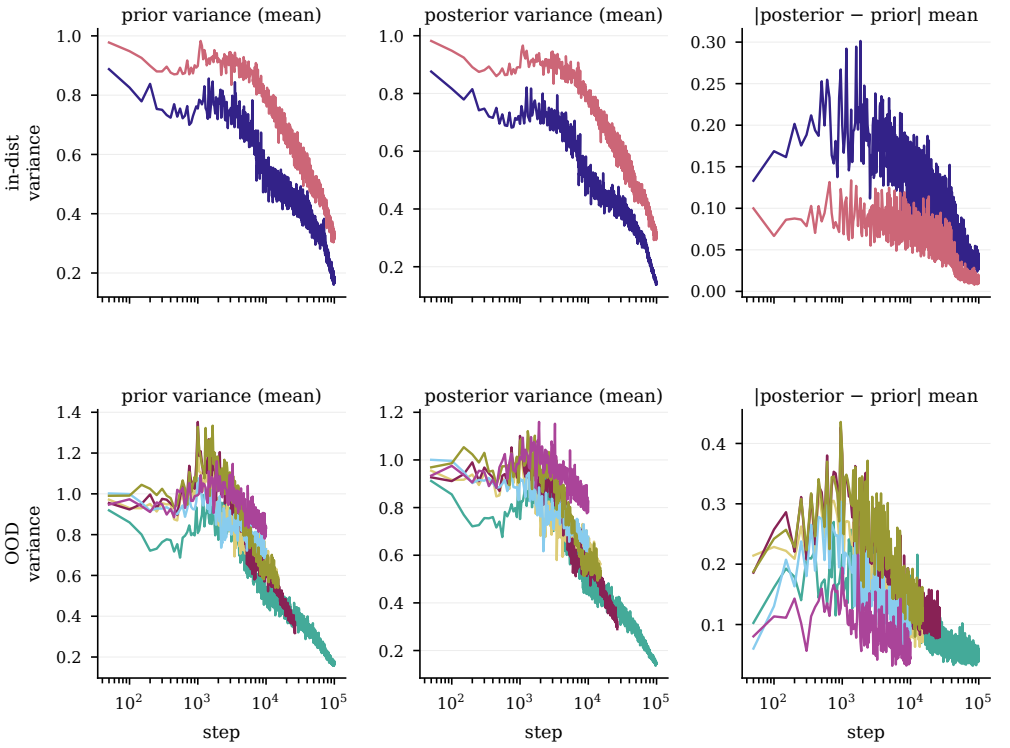
Fixed validation bank: context fit and latent use

x: optimiser step (log scale). Values as logged.



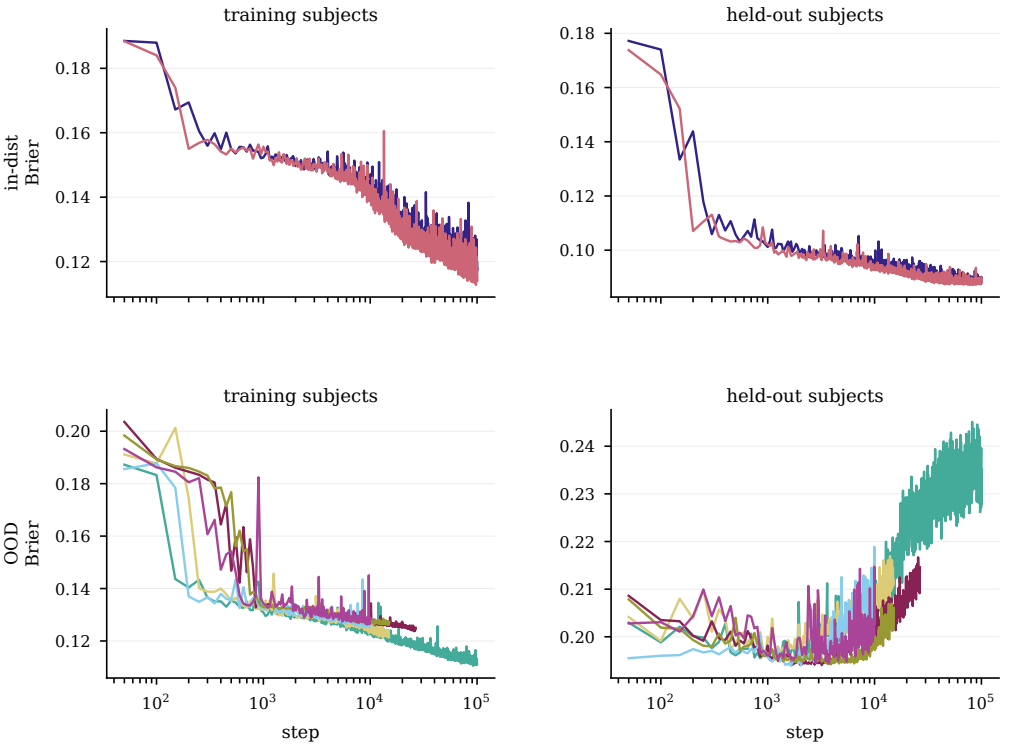
Latent distribution

x: optimiser step (log scale). Values as logged.



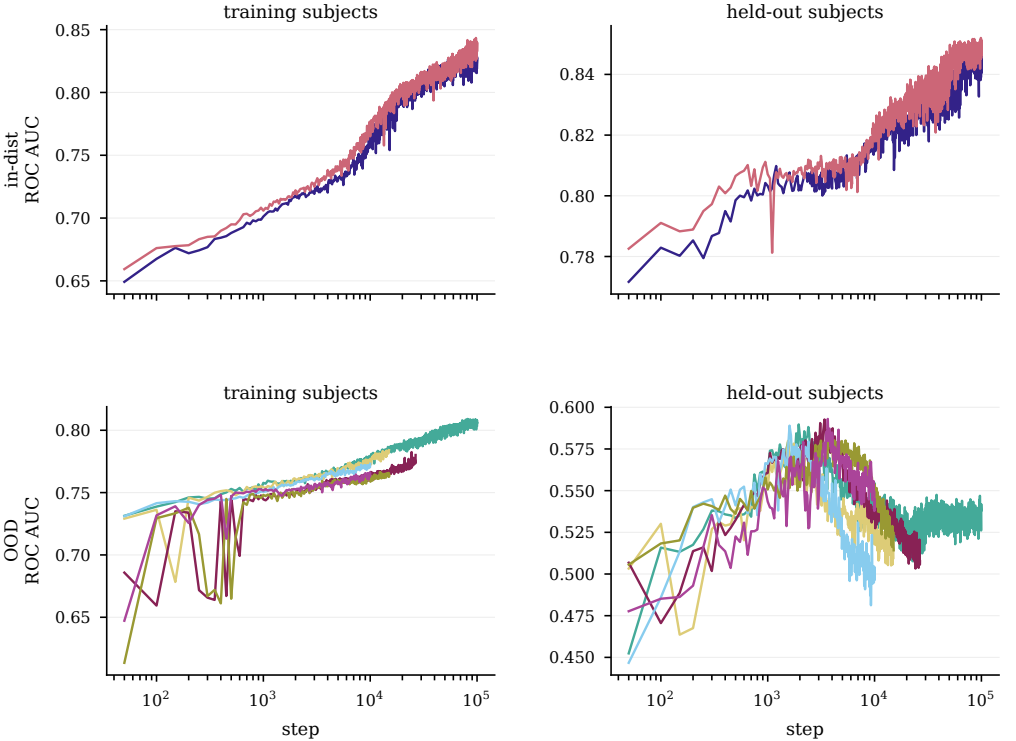
Generalisation: same-subject split-half Brier

x: optimiser step (log scale). Values as logged.



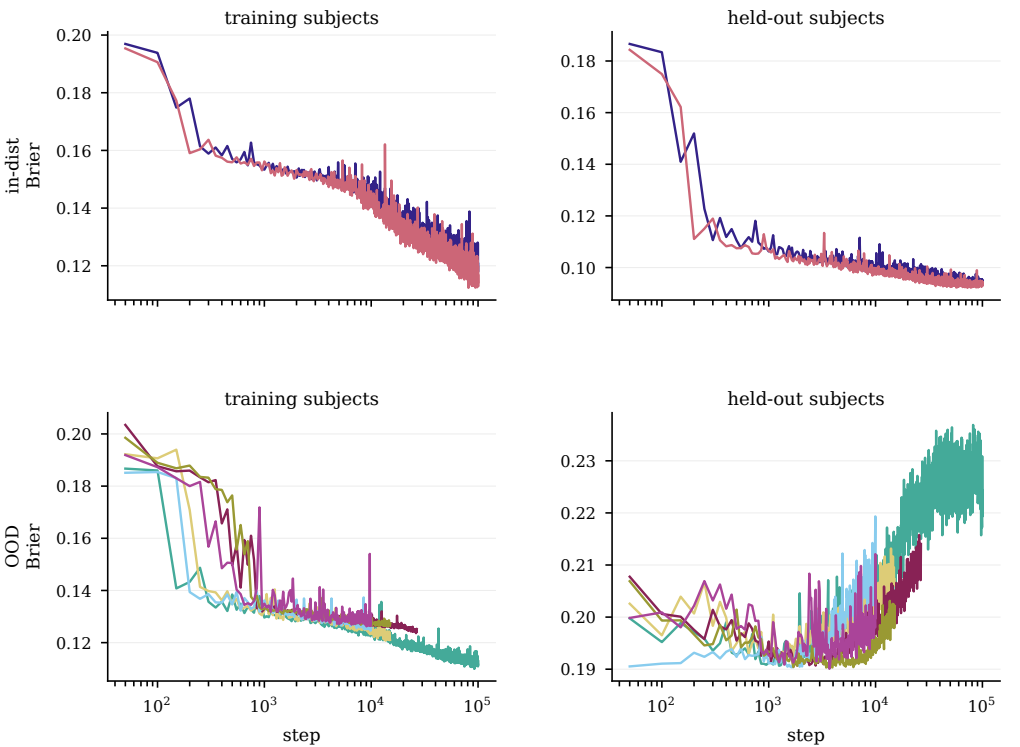
Generalisation: same-subject split-half AUC

x: optimiser step (log scale). Values as logged.



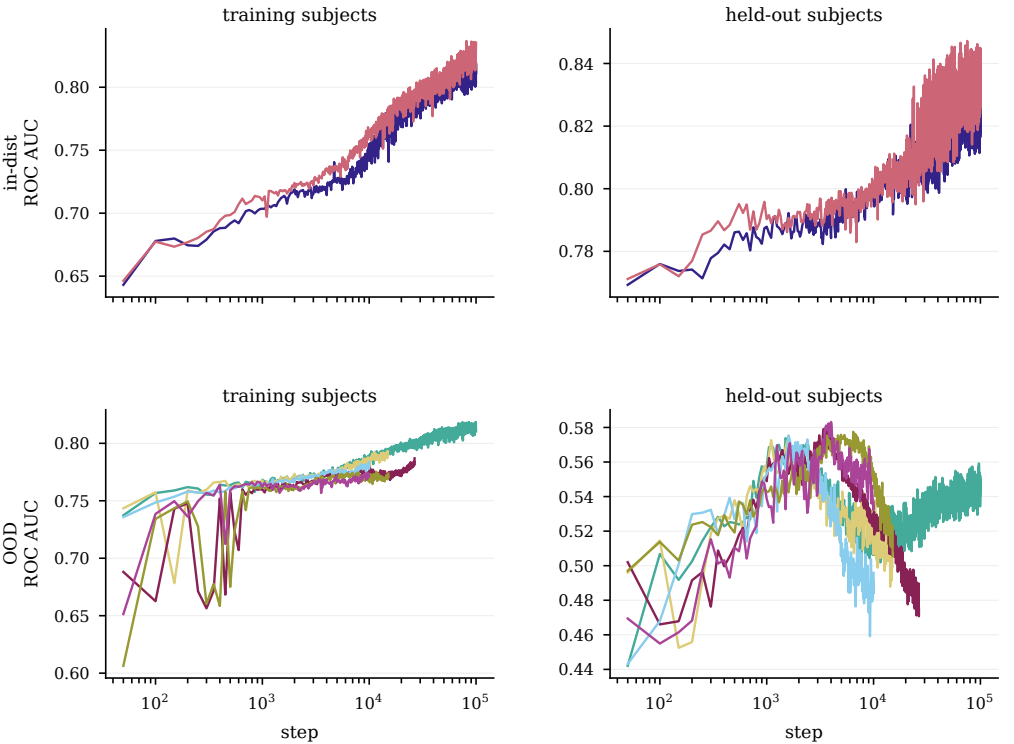
Generalisation: same-subject response-holdout Brier

x: optimiser step (log scale). Values as logged.



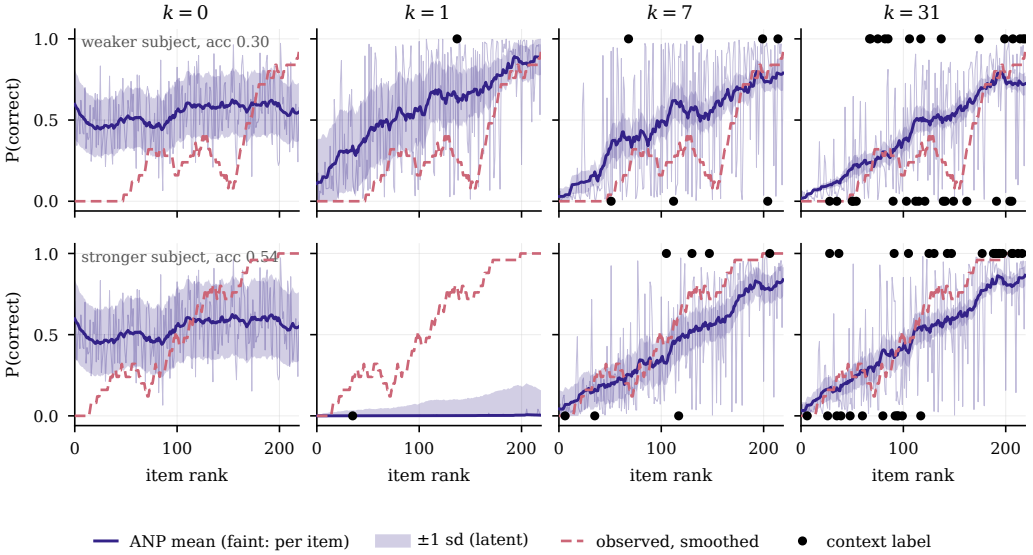
Generalisation: same-subject response-holdout AUC

x: optimiser step (log scale). Values as logged.



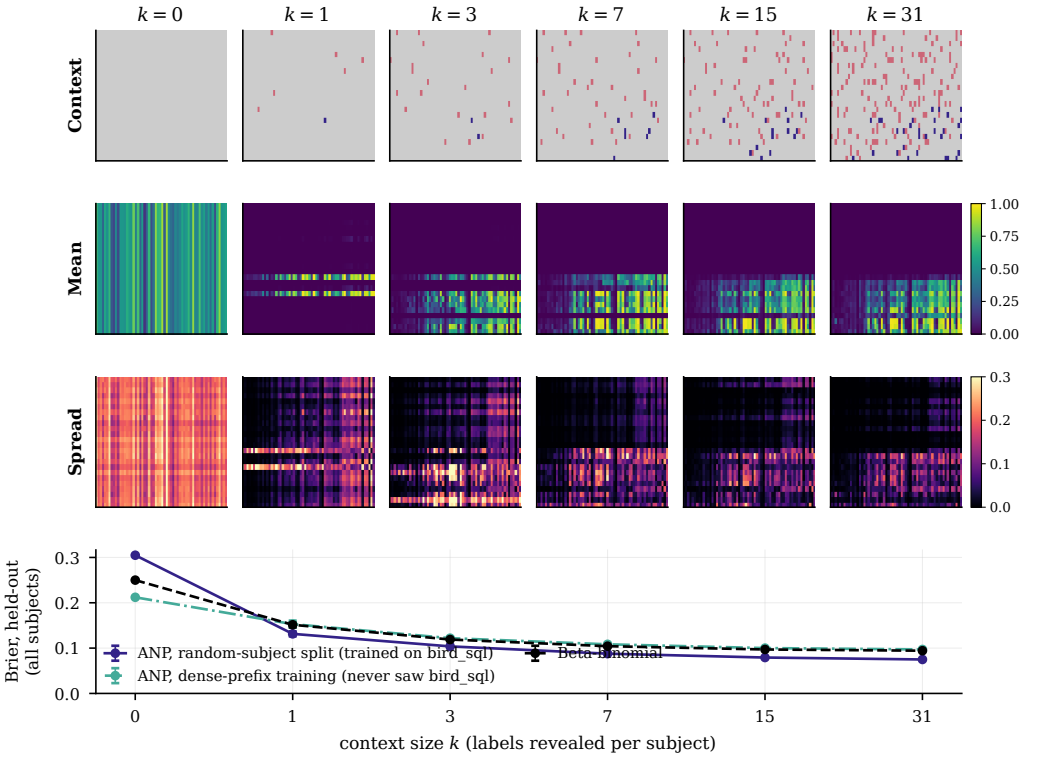
ANP learning one subject from context (in distribution)

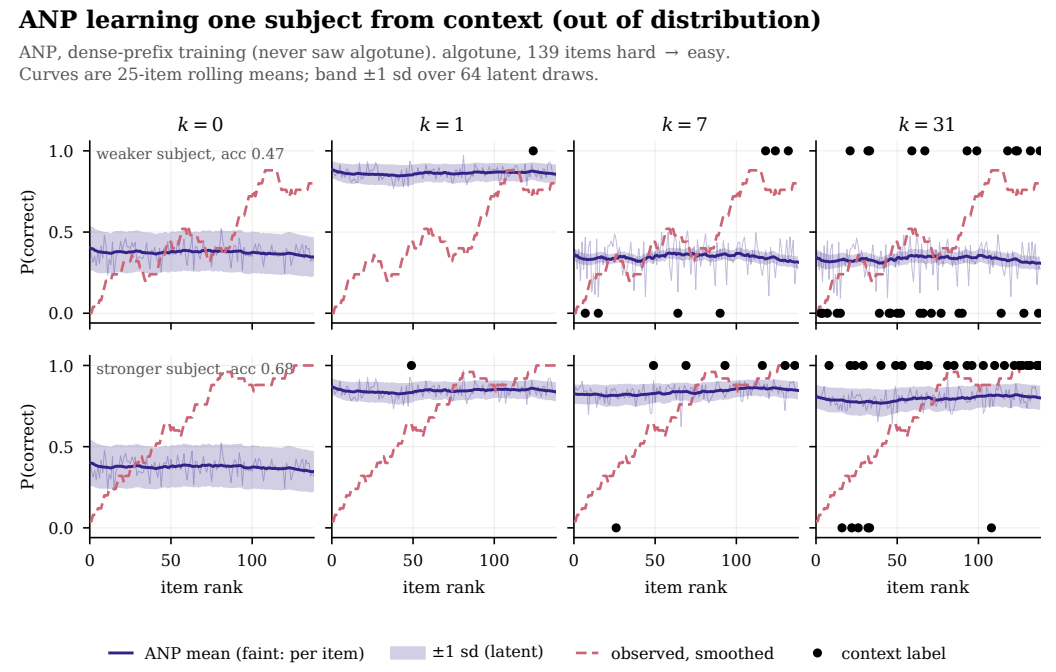
ANP, random-subject split (trained on bird_sql). bird_sql, 220 items hard → easy.
Curves are 25-item rolling means; band ±1 sd over 64 latent draws.



Response-matrix completion by the ANP (in distribution)

bird_sql: 24 subjects (weak → strong) x 60 items (hard → easy). Random context order; grey = unrevealed.




ANP learning one subject from context (out of distribution)
ANP, dense-prefix training (never saw algotune). algotune, 139 items hard → easy.
Curves are 25-item rolling means; band ±1 sd over 64 latent draws.
k = 0
k = 1
k = 7
k = 31
weaker subject, acc 0.47
stronger subject, acc 0.68
P(correct)
1.0
0.5
0.0
0
50
100
item rank
ANP mean (faint: per item)
±1 sd (latent)
observed, smoothed
context label

Response-matrix completion by the ANP (out of distribution)

algotune: 18 subjects (weak → strong) x 60 items (hard → easy). Random context order; grey = unrevealed.

